

Article

WCGAN-GA-RF: Healthcare Fraud Detection via Generative Adversarial Networks and Evolutionary Feature Selection

Junze Cai ¹, Shuhui Wu ^{2,*}, Yawen Zhang ³, Jiale Shao ² and Yuanhong Tao ³

¹ School of Computer Science and Technology, Zhejiang University of Science and Technology, Hangzhou 310023, China

² ZUST-CYU City of the Future Engineering Institute, Zhejiang University of Science and Technology, Hangzhou 310023, China

³ School of Science, Zhejiang University of Science and Technology, Hangzhou 310023, China; taoyuanhong12@zust.edu.cn (Y.T.)

* Correspondence: s.wu@zust.edu.cn or swuhzh@163.com

Abstract

Healthcare fraud poses significant risks to insurance systems, undermining both financial sustainability and equitable access to care. Accurate detection of fraudulent claims is therefore critical to ensuring the integrity of healthcare insurance operations. However, the increasing sophistication of fraud techniques and limited data availability have undermined the performance of traditional detection approaches. To address these challenges, this paper proposes WCGAN-GA-RF, an integrated fraud detection framework that synergistically combines Wasserstein Conditional Generative Adversarial Network with gradient penalty (WCGAN-GP) for synthetic data generation, genetic algorithm-based feature selection (GA-RF) for dimensionality reduction, and Random Forest (RF) for classification. The proposed framework was empirically validated on a real-world dataset of 16,000 healthcare insurance claims from a Chinese healthcare technology firm, characterized by a 16:1 class imbalance ratio (5.9% fraudulent samples) and 118 original features. Using a stratified 80/20 train–test split with results averaged over five independent runs, the WCGAN-GA-RF framework achieved a precision of $96.47 \pm 0.5\%$, a recall of $97.05 \pm 0.4\%$, and an F1-score of $96.26 \pm 0.4\%$. Notably, the GA-RF component achieved a 65% feature reduction (from 80 to 28 features) while maintaining competitive detection accuracy. Comparative experiments demonstrate that the proposed approach outperforms conventional oversampling methods, including Random Oversampling (ROS), Synthetic Minority Oversampling Technique (SMOTE), and Adaptive Synthetic Sampling (ADASYN), particularly in handling high-dimensional, severely imbalanced healthcare fraud data.

Keywords: healthcare fraud detection; WCGAN-GP; evolutionary feature selection; class imbalance; imbalanced learning; oversampling



Academic Editor: Muneer Ahmad

Received: 9 February 2026

Revised: 8 March 2026

Accepted: 15 March 2026

Published: 24 March 2026

Copyright: © 2026 by the authors.

Licensee MDPI, Basel, Switzerland.

This article is an open access article distributed under the terms and conditions of the [Creative Commons Attribution \(CC BY\)](https://creativecommons.org/licenses/by/4.0/) license.

1. Introduction

Healthcare fraud is a persistent and multifaceted issue for healthcare systems world-wide, resulting in considerable financial losses and compromised equity in service delivery [1]. The financial implications are substantial. For instance, China's healthcare spending reached RMB 23 trillion in 2021 (6.5% of GDP), with approximately 10% lost to false claims [2]. The US estimates annual losses of USD 120–410 billion [3], and comparable loss ratios have been reported across other major economies, underscoring the urgent need for advanced fraud detection methods globally.

Effective fraud detection requires sophisticated data-driven approaches capable of analyzing large-scale medical data and identifying subtle fraud patterns [1]. The theoretical foundation of fraud detection draws from three interconnected research streams: (1) **anomaly detection theory**, which posits that fraudulent transactions exhibit statistical deviations from normal behavioral patterns [4]; (2) **imbalanced learning theory**, which addresses the fundamental challenge that minority class samples (fraud) are insufficient to train robust classifiers in high-dimensional spaces [5]; and (3) **feature selection theory**, which establishes that discriminative feature identification is critical for separating fraudulent from legitimate claims in sparse data regimes [6]. These theoretical perspectives inform our methodological choices. Generative augmentation addresses the data scarcity problem identified by imbalanced learning theory, while evolutionary feature selection operationalizes feature selection theory to identify fraud-discriminative attributes.

Researchers have developed various techniques for fraud detection, ranging from rule-based systems to machine learning approaches. However, conventional detection approaches face two critical challenges. First, fraud techniques have become increasingly sophisticated—perpetrators continuously evolve their strategies to evade detection, employing complex billing schemes and collusion networks that closely mimic legitimate claims. These evolving fraud patterns render rule-based systems and static models ineffective, as they cannot capture the dynamic and adversarial nature of fraudulent activities. Second, the availability of labeled fraud data is severely limited. Fraudulent cases constitute only a small fraction of healthcare claims (typically $< 10\%$), resulting in extreme class imbalance that biases conventional classifiers toward the majority class. Moreover, the high dimensionality of claim data (often > 100 features) compounds this problem, as sparse minority samples are insufficient to train robust models across the vast feature space. From a theoretical standpoint, these challenges can be framed as follows: the anomaly detection problem is compounded by concept drift (evolving fraud tactics), while the imbalanced learning problem is exacerbated by the curse of dimensionality in high-dimensional feature spaces [7]. These dual challenges (sophisticated, evolving fraud tactics and limited, imbalanced training data) necessitate advanced data-driven approaches capable of both learning from scarce fraudulent examples and adapting to emerging fraud patterns.

Advancements in machine learning and deep learning techniques offer promising solutions to these challenges. Machine learning algorithms, such as the support vector machine with evolutionary algorithms [8], the adaptive learning and Deep Learning approach with an association engine [9], GAN with XGBoost [10], graph neural networks with reinforcement learning [11], and CGAN [12] have shown significant potential to improve the performance of machine learning models. One of the most popular approaches in the literature is GAN. GANs can generate realistic samples that conform to the distribution of the original data, which allows them to excel over most algorithms, especially in detection problems. However, GANs could result in mode collapse and vanishing gradients. Therefore, using feature engineering can help address this issue. Crucially, existing approaches typically address either data imbalance or feature selection in isolation, leaving a theoretical gap on how generative augmentation and feature selection interact to jointly improve classification in imbalanced, high-dimensional spaces.

Motivated by this gap, we propose an integrated HFD framework that synergistically combines WCGAN-GP-based synthetic data generation, GA-RF-based feature selection, and ensemble classification. Our framework is grounded in the theoretical premise that addressing data imbalance and dimensionality reduction simultaneously creates a virtuous cycle in which high-quality synthetic samples enable more robust feature importance estimation, while discriminative feature selection reduces noise in the generative training process. Specifically, we investigate three key aspects: (1) Whether WCGAN-GP can

effectively augment minority class samples while preserving distributional characteristics of fraudulent claims. (2) Whether GA-based feature selection can identify discriminative feature subsets that outperform conventional dimensionality reduction methods (e.g., PCA, Lasso). (3) Whether the synergistic combination of generative augmentation, evolutionary feature selection, and ensemble classification yields superior performance compared to each component operating independently.

To our knowledge, this work represents the first successful implementation of WCGAN-GP in healthcare fraud detection. Our contributions address the theoretical and practical limitations of existing approaches through three complementary innovations:

1. **Distribution-preserving generative augmentation for extreme class imbalance.** We propose a WCGAN-GP-based generative model to address the challenge of scarce fraudulent samples in healthcare claims data. Analytically, this contribution is significant because unlike conventional oversampling methods that perform local interpolation between existing minority instances (e.g., SMOTE, ADASYN), WCGAN-GP learns the underlying probability distribution of fraudulent claims and generates diverse, realistic synthetic samples that capture the full complexity of fraud patterns. The theoretical advantage is twofold. The Wasserstein distance provides meaningful gradients even when real and generated distributions have disjointed supports—a critical advantage when modeling rare fraudulent behaviors that occupy distinct regions in the feature space. The gradient penalty term ensures training stability and prevents mode collapse, enabling robust synthesis of complex, high-dimensional tabular healthcare data.
2. **Evolutionary feature selection for high-dimensional sparse data.** We develop a GA-Random Forest (GA-RF) feature selection module to address the curse of dimensionality in imbalanced healthcare datasets, where limited minority samples cannot support robust model training in high-dimensional feature spaces. This contribution is analytically distinct from prior work because unlike filter-based methods (e.g., correlation thresholds) that ignore feature interactions, or linear transformation approaches (e.g., PCA) that sacrifice interpretability, GA-RF employs Random Forest as the fitness evaluator to assess candidate feature subsets, directly optimizing for classification performance while efficiently navigating the combinatorial feature space through evolutionary search. The practical implication is that this approach captures nonlinear feature interactions and identifies the most discriminative feature subsets, reducing the feature space by 65% (from 80 to 28 features) while maintaining competitive detection accuracy.
3. **Comprehensive empirical validation with real-world data.** We conduct rigorous experimentation to validate the proposed framework on a real-world healthcare claims dataset (16,000 samples, 16:1 imbalance ratio, and 118 features). The analytical contribution lies in the validation protocol, which includes (1) stratified train-test split with five-fold repeated experiments to ensure reproducibility; (2) comprehensive comparison with conventional oversampling baselines (ROS, SMOTE, ADASYN); and (3) ablation studies to quantify the contribution of each component. These experiments empirically verify our theoretical premise that the synergistic combination of WCGAN-GP and GA-RF yields significant improvements in precision ($96.47 \pm 0.5\%$), recall ($97.05 \pm 0.4\%$), and F1-score ($96.26 \pm 0.4\%$), with GA-RF achieving 65% feature reduction while maintaining competitive accuracy. To our knowledge, this work represents the first successful application of WCGAN-GP in healthcare fraud detection.

The remainder of this paper is organized as follows: Section 2 reviews related work on machine learning in HFD, data sampling methods, and genetic algorithms. Section 3 presents theoretical preliminaries of generative adversarial networks. Section 4 details

the proposed WCGAN-GA-RF architecture and training process. Section 5 presents experimental validation, including dataset description, ablation studies, and performance comparisons. Section 6 concludes with limitations and future research directions.

2. Related Work

2.1. Machine Learning in HFD

At its core, HFD seeks to reliably distinguish deceptive transactions from legitimate medical activities, a crucial step toward protecting both the financial health of insurers and the quality of patient care. The healthcare landscape is a complex, interactive system among patients, their providers, and payers, all having motivations and limitations of their own. Finding fraudulent transactions and separating them from genuine medical actions is the key to discovering healthcare fraud. A sophisticated network of clients, providers, and users interact with the medical program. It is crucial to ensure the economic and individual attention of healthcare care companies. Each of these teams has its own limitations and arguments. Cheating does not usually look like clear bad habits in a difficult program. Rather, it often hides in simple-looking records, making it difficult to find, such as trying to find a needle in a haystack. Machine learning has come a long way in this regard. These techniques can examine large amounts of data to uncover simple, hidden patterns that people may miss when using conventional quantitative methods or hand-checking. It is hardly surprising that recent studies on how to use equipment understanding to find and prevent healthcare fraud have increased tremendously.

Academics generally employ unsupervised and supervised learning as their main methods to determine fraud. When there are unlabeled data, the unsupervised method has been very valuable for detecting unusual patterns. In this area, properly established methods, such as K-means clustering [13–16], Local Outlier Factor Analysis [17], and Principal Component Analysis [18], have established a strong foundation. The advanced types of unsupervised learning resources have recently gained more interest. These include deep learning techniques, such as LSTM networks [19,20] and Graph Neural Networks [21] as well as a specific neuronal network designed for operating without labeled files, such as Autoencoders [22,23]. All of these approaches are becoming more adept at offering increasingly sophisticated detection capabilities.

On the other hand, supervised methods connect insight characteristics to known fraud cases using effectively labeled data. Popular methods like Gradient-boosted Decision Trees, Random Forests, and plain Decision Trees are included in this class. K-nearest neighbors [24–27] and various neural network approaches, as well as more developed methods like Deep Neural Networks [28], are other important supervised machine learning methods. In addition to strengthening supervised learning, deep learning is continuing to expand. Then, Convolutional Neural Networks [29–31] are generally employed to improve the precision of the monitoring.

The development of hybrid techniques that combine different methods is one of the most amazing things. These combined methods may be the best course of action, according to the latest research combining Autoencoders with Neural Networks [32] or incorporating tree-based algorithms with other techniques [33]. They may significantly improve the ability to identify fraudulent actions in a variety of situations.

A significant challenge in fraud detection is the limited availability of labeled data, coupled with the intrinsic class imbalance in the datasets, because there are substantially more legitimate instances than fraudulent activity ones. It is difficult to effectively train machine learning models because there are not enough real-world examples of fraud. In addition, having enough fair divisions and a fine sample size when training these samples depends on how efficient they are in the training process. Methods like WCGAN-GP and

feature engineering are important in overcoming these issues. These methods improve the functioning of fraud detection designs by making great fake information and stability data.

Although existing fraud detection systems already have a solid data-driven foundation, enduring challenges such as inadequate datasets and skewed class distributions continue to call for more sophisticated solutions. Bringing together advanced approaches, particularly synthetic data generation through techniques such as GANs and refined feature engineering, appears promising to raise both the accuracy and trustworthiness of HFD.

2.2. Data Sampling Methods in HFD

A persistent issue in the detection of healthcare fraud is the pronounced imbalance in datasets, where legitimate medical claims dramatically outnumber confirmed cases of fraud. Due to uneven distribution, it is hard to establish reliable machine learning models. To address class imbalance, academics typically employ four strategies: (A) oversampling in the minority class group, (B) undersampling to refine majority class examples, (C) adjusting the weight of each instance in their loss functions, and (D) generating synthetic samples using GANs. Each of these approaches has advantages and disadvantages. Generally, how effective they are depending significantly on the nature of the dataset and the specific fraud detection scenario at hand.

(A) *Oversampling*: Oversampling strategies seek to correct uneven class distributions by improving the presence of minority categories in training data. Random Oversampling (ROS), one of the most common methods, simply duplicates existing minority-class samples. More advanced synthetic oversampling techniques generate new instances by algorithmically interpolating between observed minority samples. The Synthetic Minority Oversampling Technique (SMOTE) and its variations, such as Borderline SMOTE and Adaptive Synthetic Sampling, are widely used approaches by strategically interpolating between existing examples [34–36]. In HFD systems, these techniques are frequently employed. However, it is important to note that oversampling carries certain risks. It can either present unrepresentative synthetic cases or cause model overfitting, especially when the created samples do not capture the real pattern in the underlying data.

(B) *Undersampling*: Undersampling addresses class imbalance by reducing instances from the majority class. While likely reducing computational costs, it may also result in the loss of valuable data and compromise model generalization. Methods such as BalanceCascade [37] and EasyEnsemble [38] have been introduced to address this problem. Both techniques have been demonstrated as effective in preserving information while reducing the majority class samples in domains such as healthcare.

(C) *Weighting*: An alternative method for solving class imbalance is by focusing more on minority class samples in the training [39–41]. This is either by adjusting the weights of classes inside the loss function or by designing loss functions. Nevertheless, one of the main issues is that they are sensitive to exact hyperparameter choices, which in most cases require a lot of tuning. Even in scenarios of extreme imbalances, weighting approaches might not be effective enough since they fail to expand the literal amount of minority examples. In its turn, this has been answered by some recent research into learning in transformed high-dimensional spaces, where the role of a better representation of features can be utilized by allowing models to capture patterns using less data on minority classes.

(D) *GANs-Based Methods*: GANs have been found to have potential in solving the issue of class imbalance by generating synthetic samples of the minority class. In HFD, GAN-based methods are used to create realistic cases of fraud to augment real-world data to optimize the performance of supervised models. As an illustration, the structured Conditional Tabular GAN has been used to over-sample minority classes data, which causes improved results with classifiers [12]. Another method uses GANs to produce

synthetic samples labeled with anomaly scores, which are then used to train supervised detection systems [42]. Despite these advantages, GAN-based approaches face several challenges in healthcare fraud detection, including (1) training instability and mode collapse when modeling complex, high-dimensional tabular data; (2) difficulty in capturing the full distribution of rare fraudulent patterns; and (3) risk of generating low-quality synthetic samples that degrade classifier performance. These challenges motivate our adoption of WCGAN-GP with Wasserstein distance and gradient penalty mechanisms.

In addition to solving the issue of class imbalance, the reduction in dimensionality is a key factor in improving the effectiveness and efficiency of fraud diagnosis. Methods such as Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA), and Quadratic Discriminant Analysis (QDA) reduce the size of feature spaces and retain important patterns of discrimination. The other direction of research is the choice of significant features. The important indicators of fraud are located using different techniques, among which are greedy search techniques and spectral-based selection techniques. Dimensionality reduction can be used together with semantic embedding and other sophisticated feature engineering techniques to build a more thorough analytical system that further improves the performance of the models.

2.3. Genetic Algorithms in HFD

Previous methods for accessing healthcare insurance fraud tended to involve a mixture of many computational structures. One such study [43] combined a genetic algorithm and a K-Nearest Neighbor (KNN) classifier to study the professional profiles of general practitioners. In this approach, the genetic algorithm method was applied to identify the best feature weights, which in turn improved KNN. The model is applicable in the detection of healthcare fraud, as it has performed excellently in modeling activities. It developed another impactful study [44] in which it proposed a fuzzy clustering method with the Salp Swarm Algorithm (SSA) to calculate the best cluster centers and eliminate outliers by undersampling. Through these studies, it is clear that the mixture of different approaches would be a great boost in fraud detection.

As the healthcare sector expanded, it faced a parallel increase in fraudulent activities, creating significant challenges for its healthy development. In response, academics in [45] developed an improved adaptive genetic algorithm (IAGA) coupled with a back-propagation neural network. This IAGA-BP framework was developed to enhance the network's starting weight options. It helps in solving well-known issues, such as convergence to local minima, slow training speed, and sensitivity to specific data samples. This process was tested more quickly and accurately than traditional genetic techniques using historical automobile insurance claims as a test. It becomes more difficult to detect fraud because of the complex relationships between doctors, patients, and companies in the medical industry. Another study [8] adopted a dynamic weighted search space strategy based on an evolutionary algorithm, hybridized with a support vector machine using a modified whale optimization algorithm (MWOA). This approach was rigorously tested with several evaluation metrics. The strategy first extracted anomalies from the information before using powerful tools like CATBoost (<https://catboost.ai>), XGBoost (<https://xgboost.readthedocs.io>), scikit-learn (Random Forest, Decision Tree: <https://scikit-learn.org>), and LightGBM (<https://lightgbm.readthedocs.io>). This improved the reliability of healthcare claim predictions.

A study develops a new technique known as the Deep Emended Genetic Algorithm (DEGA) to combat the growing problem of fake online transactions [46]. By incorporating a few key new innovations, including better ways to combine and choose data, adaptable changes to various circumstances, and a system that encourages competition to find the

best options, this approach improves the ability to detect fraud. Working in tandem with DEGA, a classification system that employs dual sigmoid loss functions to better manage datasets with significantly imbalanced labels.

While these prior studies demonstrate the potential of genetic algorithms in fraud detection, they primarily employ GA for hyperparameter tuning or hybrid model optimization. In contrast, our proposed GA-RF method leverages evolutionary optimization specifically for feature subset selection in severely imbalanced, high-dimensional healthcare datasets. Unlike wrapper methods that are computationally prohibitive for large feature spaces, GA-RF directly optimizes for classification performance while efficiently exploring the combinatorial feature space. This approach is particularly suited for healthcare fraud detection where (1) the minority class (fraudulent claims) is extremely sparse, requiring discriminative feature selection to improve signal-to-noise ratio; and (2) nonlinear feature interactions are prevalent, necessitating a search strategy that can capture complex dependencies beyond simple correlation measures.

3. Preliminaries

This section establishes the theoretical foundations for our proposed framework. We present the mathematical background of generative adversarial networks and their variants, with specific emphasis on how these techniques address the challenges of healthcare fraud detection, including (1) severe class imbalance where fraudulent claims constitute less than 1% of all claims; (2) high-dimensional sparse feature spaces arising from complex medical coding systems; and (3) the need for generating realistic synthetic fraud samples that preserve the statistical properties of genuine fraud patterns. Understanding these theoretical underpinnings is essential for appreciating the design choices in our WCGAN-GA-RF framework presented in Section 4.

3.1. Generative Adversarial Networks

GANs [47] have attracted a lot of interest because they have been successful in a variety of fields and are a powerful tool for deep generative modeling. The core idea revolves around a competitive training paradigm between two neural networks: the generator G and the discriminator D . Here, G takes random noise z , which is usually sampled from a straightforward distribution P_z , such as uniform or Gaussian, and produces synthetic outputs $G(z)$ designed to resemble samples from the true data distribution P_{data} . Meanwhile, D acts as a critic, learning to differentiate actual data examples x from those produced by G . This adversarial process is captured in the following mini-max objective:

$$\min_G \max_D L(D, G) = \mathbb{E}_{x \sim P_{\text{data}}(x)} [\log D(x)] + \mathbb{E}_{z \sim P_z(z)} [\log(1 - D(G(z)))].$$

When the discriminator is trained to maximize the following expression,

$$-\mathbb{E}_{x \sim P_{\text{data}}(x)} [\log D(x)] - \mathbb{E}_{z \sim P_z(z)} [\log(1 - D(G(z)))],$$

the generator is actually trying to reduce the Jensen–Shannon divergence between the real data distribution P_{data} and its own generated data distribution P_{data} . The JSD offers a symmetric and interpretable measure of how close two distributions are, forming a key part of the theoretical basis for GANs.

The main weakness of the original GAN architecture [47] is that it does not know what kind of data it generates in particular categories. This is especially important when the data are highly imbalanced, in which case more examples of under-represented classes will be needed. Conventional GANs have a silent and strong memory of complicated patterns of data, yet they cannot guide the generation procedure to specific classes.

Due to this drawback, CGANs were created, which add instructions to the generation process. The key innovation is to provide each network with additional information, usually in the form of category labels y . The discriminator does two functions. It determines whether the samples belong to the requested group and whether the samples are realistic. This refined training becomes familiar to the generator in which he creates samples that are capable of fitting specific categories. The ability provides CGANs with a special potential to solve the problem of data imbalance using focused data augmentation [48]. CGAN is based on the original GAN in that it retains a similar dynamic training architecture but incorporates conditional components. The mathematical formulation is the following:

$$\min_G \max_D L(D, G) = \mathbb{E}_{x \sim P_{\text{data}}(x)} [\log D(x|y)] + \mathbb{E}_{z \sim P_z(z)} [\log(1 - D(G(z|y)))]$$

where the vertical bar $|$ denotes conditioning, meaning the discriminator evaluates x given the class label y , and the generator produces samples conditioned on y . This conditioning on y enables controlled generation. In practice, this means that the model not only gains the ability to produce samples of desired types, but also benefits from more reliable training dynamics compared to standard GANs [49].

3.2. Wasserstein GANs with Gradient Penalty

While the original GAN has been around for some time, it continues to face challenges that limit its unstable gradients and mode collapse. The study [50] suggests that many of these issues originate from the use of JSD to measure the similarity between distributions. One particular weakness appears when the actual and generated distributions do not overlap. In this case, the JSD stays fixed at $\log 2$, causing the generator's gradients to disappear. As a result, the generator stops receiving useful feedback and cannot be improved through training. At the same time, the generator's loss function is often influenced by what is known as reverse KL divergence, pushing it toward what [51] refers to as "safe" sampling behavior. This tendency causes the model to generate only a narrow set of plausible outputs rather than exploring the full variety of the target data, which is referred to as mode collapse.

WGAN addresses these challenges by optimizing the Earth-Mover (EM) distance (also called the Wasserstein-1 distance) instead of the JSD. When using the EM distance, the distance between a given distribution and P_{data} can be calculated even if neither of them has any overlap with the other. Therefore, the problems mentioned above can be addressed. Although calculating the EM distance is troublesome in real-world applications, the study [50] proposed that the EM distance can be approximated by changing the objective of GAN to Equation (1) as long as D is a k -Lipschitz function. This constraint can be enforced by weight clipping, which limits the weights of D to a compact interval $[-0.01, 0.01]$:

$$\min_G \max_D L(D, G) = \mathbb{E}_{x \sim P_{\text{data}}(x)} [D(x)] - \mathbb{E}_{z \sim P_z(z)} [D(G(z))]. \quad (1)$$

Although WGANs marked clear progress in the development of generative models, researchers soon observed practical drawbacks in their use of weight clipping. The study [52] indicates that this technique tends to limit what the discriminator can learn and may still cause the gradients to become negligibly small during training. In response, they designed WGAN-GP, a revised version that imposes a gradient penalty instead of reducing the weights of the network.

Intuitively, the gradient penalty works as follows: to ensure stable training, the discriminator's gradients should remain well-behaved (specifically, the norm should be close to one). If the gradients become too large or too small, the training becomes unstable. The gradient penalty term measures how far the gradient norm deviates from one and

adds a penalty to the loss function accordingly. This is analogous to adding a “smoothing” constraint that prevents the discriminator from changing too abruptly.

The selection of WGAN-GP over other GAN variants is motivated by three critical requirements in HFD: (1) **Training stability**: Healthcare claim data exhibit high dimensionality (80 features in our dataset) and severe imbalance (fraud rate < 1%), which exacerbate the training instability of standard GANs. WGAN-GP’s Wasserstein distance provides more stable gradients during training, preventing mode collapse where the generator produces limited diversity in synthetic fraud samples. (2) **Sample quality**: The gradient penalty mechanism ensures that synthetic fraud samples maintain statistical similarity to genuine fraud patterns, which is essential for effective downstream classification. (3) **Conditional generation capability**: By incorporating class labels (fraud vs. legitimate), WCGAN-GP enables targeted augmentation of the minority fraudulent class without wasting computational resources on generating majority class samples that are already abundant.

Inspired by these outcomes, we leverage the WCGAN-GP setup to handle imbalanced data in HFD. The method optimizes the following objective:

$$\min_G \max_D L(D, G) = -\mathbb{E}_{x \sim P_{\text{data}}(x)}[D(x)] + \mathbb{E}_{z \sim P_z(z)}[D(G(z))] \quad (2)$$

$$+ \lambda \cdot \mathbb{E}_{\hat{x}} \left[(\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1)^2 \right] \quad (3)$$

where $\hat{x}$ is the linear interpolation between real and generated data samples, while λ is the gradient penalty coefficient (implemented as gamma in the program code), $\nabla_{\hat{x}} D(\hat{x})$ is the discriminator gradient with respect to the interpolated data points. The gradient penalty is computationally implemented by incorporating a regularization term into the discriminator’s loss function, which penalizes the norm of the discriminator’s output gradient.

This section has presented the theoretical foundations of GANs, CGANs, and WGAN-GP. The evolution from basic GANs to conditional variants addresses the need for targeted class augmentation, while the Wasserstein distance with gradient penalty resolves training instability issues. These theoretical advancements provide the mathematical basis for our WCGAN-GP component in Section 4, which combines conditional generation with stable training dynamics to effectively augment minority fraud samples in healthcare claims data.

3.3. Theoretical Framework for Component Interaction

This subsection articulates the theoretical framework explaining how the three components of our WCGAN-GA-RF framework—generative augmentation (WCGAN-GP), evolutionary feature selection (GA-RF), and ensemble classification (RF)—interact synergistically to address the challenges of imbalanced, high-dimensional healthcare fraud detection. We formalize this interaction through testable hypotheses grounded in learning theory.

The three components form a cascading pipeline with bidirectional benefits:

1. WCGAN-GP → GA-RF: Generative augmentation creates a more balanced training distribution, enabling more robust estimation of feature importance. In severely imbalanced settings (<6% fraud rate), standard feature importance measures are biased toward features that discriminate the majority class. By augmenting minority class samples, WCGAN-GP allows the Random Forest fitness evaluator in GA-RF to identify features that are genuinely discriminative for fraud detection, rather than features that merely distinguish the abundant legitimate claims.
2. GA-RF → RF Classifier: Feature selection reduces noise and computational burden while preserving discriminative power. The selected 28-feature subset captures the essential fraud patterns identified through evolutionary search, enabling the Random Forest classifier to train more efficiently and generalize better to unseen fraud cases.

3. **Implicit feedback loop:** The quality of synthetic samples generated by WCGAN-GP is indirectly validated through the downstream classification performance. If synthetic samples introduce artifacts or noise, GA-RF will struggle to identify consistent discriminative features, and the final RF classifier will exhibit degraded performance on the test set. This implicit feedback mechanism provides a quality control signal for the generative process.

According to this theoretical construct, we suggest the following empirically testable hypotheses that inform our experimental validation:

- **H1 (Generative Augmentation Effectiveness):** WCGAN-GP-generated synthetic samples achieve higher classification performance than methods of interpolation-based oversampling (SMOTE, ADASYN) since Wasserstein distance learning is able to fit the entire distribution of fraudulent claims as opposed to local interpolation between samples.

Prediction: WCGAN-GP obtains greater F1-scores on the minority group than SMOTE and ADASYN, especially in high-dimensional feature spaces where the minority group is in a sparse form.

- **H2 (Feature Selection Synergy):** GA-based feature selection identifies more discriminative feature subsets when applied to WCGAN-GP-augmented data compared to original imbalanced data, because balanced training data enable unbiased feature importance estimation.

Prediction: The feature subset selected by GA-RF on WCGAN-GP-augmented data have higher classification accuracy using fewer features than individual applications of GA-RF on original imbalanced data or on the SMOTE/ADASYN-augmented data.

- **H3 (Component Synergy):** The end-to-end WCGAN-GA-RF pipeline is more effective than ablated systems in which various components are ablated or substituted, demonstrating the critical nature of the synergistic combination.

Prediction: (1) WCGAN-GA-RF (full) > WCGAN + RF (no feature selection); (2) WCGAN-GA-RF > GA-RF + SMOTE (generative augmentation replaced); (3) WCGAN-GA-RF > Baseline (no augmentation, no feature selection).

- **H4 (Amplified Advantage under Severe Imbalance):** The performance advantage of WCGAN-GA-RF over baseline methods will become more pronounced as class imbalance severity increases, because generative augmentation becomes increasingly valuable when minority class samples are critically scarce.

Prediction: The performance gap between WCGAN-GA-RF and SMOTE/ADASYN is larger at 16:1 imbalance ratio than at 4:1 or 8:1 imbalance ratio.

These hypotheses are grounded in established learning theory. H1 follows from the theoretical advantage of distribution learning over interpolation. Wasserstein distance provides meaningful gradients for learning complex, multi-modal distributions of fraudulent claims. H2 is justified by the observation that feature importance estimates are unreliable when minority class samples are sparse [5]. H3 captures the synergistic interaction posited by our framework. Each component addresses a distinct challenge, and their combination creates a virtuous cycle. H4 stems from the curse of dimensionality. As imbalance severity increases, the minority class occupies an increasingly sparse region of the feature space, making generative augmentation more valuable than local interpolation methods.

4. Proposed Framework

4.1. Framework Overview

Figure 1 illustrates the complete workflow of our proposed WCGAN-GA-RF framework. The system operates in three sequential stages: (1) The WCGAN-GP component takes Gaus-

sian noise z and class labels y as input to generate synthetic minority class samples, which are combined with the original training data to form a balanced dataset. (2) Feature Selection: The GA-RF module employs evolutionary optimization to identify the most discriminative feature subset from the augmented data, reducing dimensionality from 80 to 28 features while preserving classification performance. (3) Classification: The selected features are used to train a Random Forest classifier, which makes final predictions on the test set. The dotted lines indicate data flow, while solid lines represent model training processes.

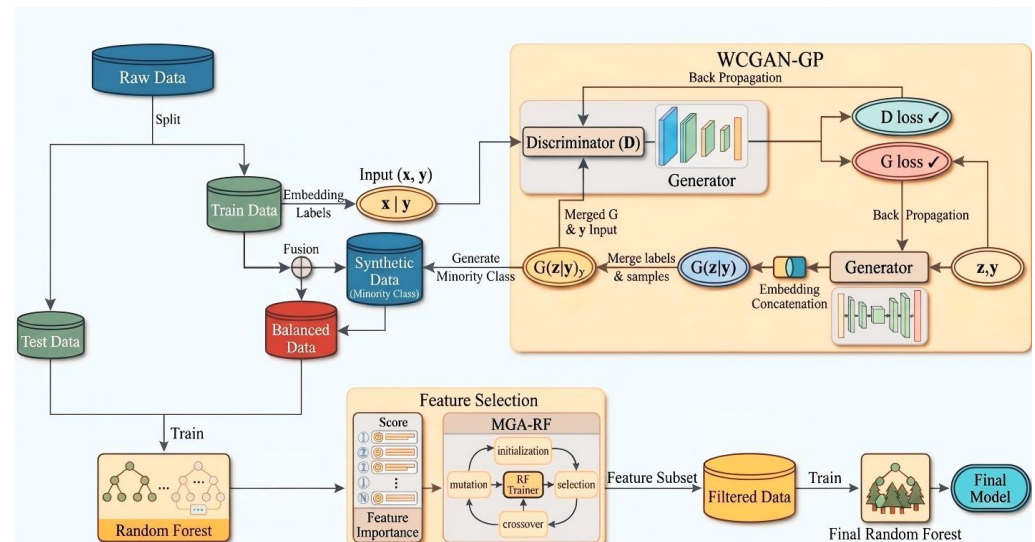


Figure 1. The proposed WCGAN-GA-RF framework operates in three sequential stages: (1) WCGAN-GP generates synthetic minority class samples to address data imbalance; (2) GA-RF identifies the most discriminative feature subset through evolutionary search; and (3) Random Forest classifier makes final fraud predictions using the selected features.

The dataset is first stratified and split into training (80%) and test (20%) sets, ensuring that minority class distribution is preserved in both splits. Data augmentation, feature selection, and model training are performed exclusively on the training set. The test set remains completely held out until final evaluation to prevent data leakage.

4.2. WCGAN-GP Architecture

The WCGAN-GP framework serves as the foundational component within the fraud detection system, specifically engineered to produce quality-enhanced synthetic data samples to mitigate the challenges of dataset imbalance. The architectural design incorporates neural network-based implementations for the generative and discriminative components, capitalizing on their efficacy in feature extraction and pattern recognition tasks. The generative module is structured with four convolutional layers incorporating ReLU activation functions, complemented by dual fully connected layers responsible for synthesizing artificial data instances. The discriminative component in parallel has a similar structural arrangement, but incorporates a sigmoid activation layer to support binary classification functions. The architectural layout enables mining complex data patterns and multi-level feature representation, which is imperative in making a valid distinction between valid and fraudulent financial activities. The generator and discriminator/critic architectures are detailed in Table 1.

We propose a solution to the issue of data imbalance in the form of CGAN networks. In the CGAN, the generator and the discriminator are trained on additional information about class labels. This conditioning information can ensure that the generator generates samples that are consistent with this additional information, whereas the discriminator can determine the authenticity of the samples depending on the input data and the condition.

CGAN presents contingent input to control the augmentation process. This is a monitored technique that enables CGAN to control the discrepancy between positive and negative data samples. CGAN guaranties the production of high-quality samples with a particular category or attribute feature.

Table 1. Generator and discriminator/critic architecture.

Layers	Generator		Discriminator/Critic	
	(Noise Dim + Label)	—	(n + Label)	—
Hidden layer1	128	Leaky-ReLU	128	Leaky-ReLU
Hidden layer2	64	Leaky-ReLU	64	Leaky-ReLU
Output layer	n	Tanh	1	Linear

Note: n denotes the number of features in the dataset (after preprocessing, $n = 80$), and Noise dim is the dimension of the random noise vector z (set to 100). Label refers to the class label for conditional generation.

These training processes are essential for generating high-quality synthetic fraud samples in healthcare claims. During generator training (Figure 2), the network learns to transform random noise into realistic healthcare claim patterns that resemble genuine fraud cases, conditioned on minority class labels to address the severe imbalance in fraud detection datasets. The discriminator training (Figure 3) ensures that only convincing synthetic samples are produced by continuously challenging the generator to improve. This adversarial dynamic is critical for healthcare fraud detection because (1) genuine fraud patterns are rare and difficult to obtain, making synthetic augmentation necessary; (2) the quality of synthetic samples directly impacts downstream classification performance; and (3) conditional generation ensures that the augmented samples belong to the underrepresented fraudulent class rather than the majority legitimate class.

Figure 2 shows that the input layer of the generator comprises a Gaussian random noise vector z and a fault label y . The label y offers auxiliary information to regulate the formation of data of particular classes. This method makes it easy to augment the data by allowing training of single models, regardless of the number of fault classes. The input sequence undergoes two different layers of multiplication before being doubled in size. The initial neural network configuration comprises 16 units and 16 feature maps, whereas the subsequent neural network configuration comprises 8 units and a single feature map. The discriminator consists of two one-dimensional neural network layers featuring 8 and 16 neurons, respectively, arranged sequentially following the input layer, as shown in Figure 3. The feature maps are flattened to form a fully connected layer. The output layer subsequently provides two prediction labels: data source and fault class.

A significant challenge in training most generative adversarial models is mode collapse, where the generated samples lack diversity. To address this, we adopt the WCGAN-GP loss function (Equation (2)) with the following theoretical justification:

- Wasserstein distance (L_{WD}): Provides meaningful gradients even when real and generated distributions have disjoint supports—critical for fraud detection where the minority class occupies a sparse region in the 80-dimensional feature space.
- Gradient penalty (L_{GP}): Enforces the 1-Lipschitz constraint necessary for Wasserstein distance estimation. The penalty $\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1)^2$ prevents discriminator gradients from becoming too sharp (causing mode collapse) or too flat (causing gradient vanishing), which is essential for tabular healthcare data with varying feature scales.

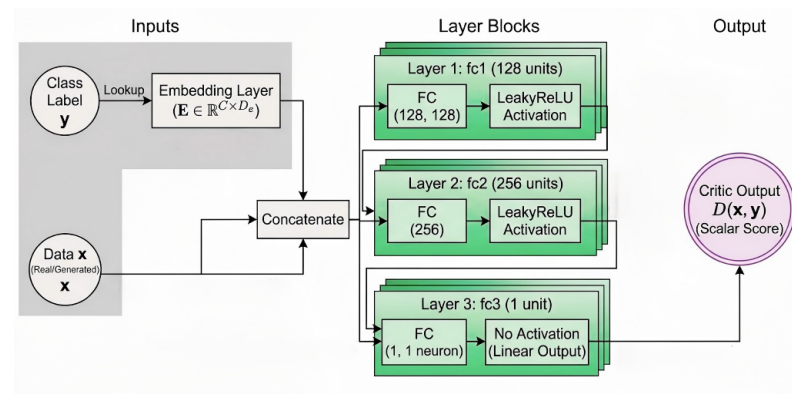


Figure 2. The generator learns to produce realistic synthetic fraud samples by transforming random noise into data that can fool the discriminator, guided by class labels to ensure targeted minority class augmentation.

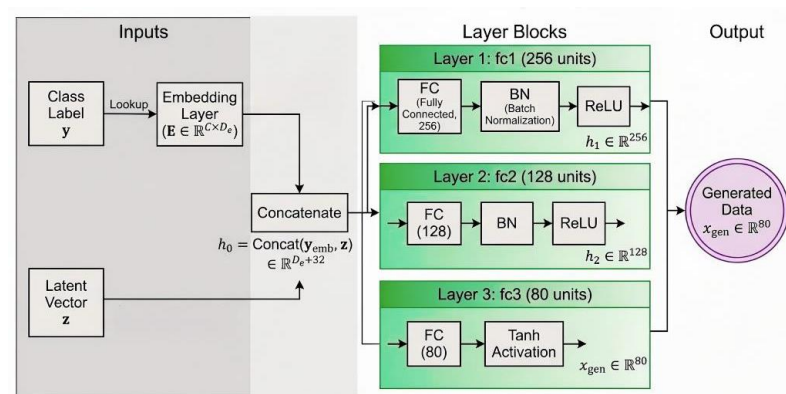


Figure 3. The discriminator learns to distinguish between genuine healthcare claims and synthetic samples, providing gradient feedback to improve the generator's ability to produce realistic fraud patterns.

The gradient penalty coefficient λ is set to 10, following the theoretical recommendation of Gulrajani et al. [52], who proved that this value satisfies the Lipschitz constraint with high probability. We empirically validated this choice through preliminary experiments with $\lambda \in \{1, 5, 10, 20\}$ on a held-out validation set, measuring (1) training stability (variance of generator loss), (2) mode coverage (number of distinct fraud patterns generated), and (3) downstream classification performance. $\lambda = 10$ achieved the optimal trade-off, with stable gradients and maximum pattern diversity. The classification weight is implicitly set to 1.0 to ensure equal emphasis on realism and conditional generation.

Such equations have a structural similarity to the loss function used in CGAN. The major difference is that it substitutes the initial binary cross-entropy loss with the Wasserstein distance loss, following the addition of a gradient penalty loss term to enforce the Lipschitz constraint [53–55].

To ensure the stability of the WCGAN-GP training, several optimization strategies are implemented as follows.

- (A) Batch Normalization: Batch normalization is used in both the generator and the discriminator and helps resolve problems with starting points, causes gradient propagation across the layers, and stops the generator from collapsing to a single output mode.
- (B) Activation Functions:
 - (i) In the output layer of the generator, the Tanh function maps the inputs to the range $[-1, 1]$. This function exhibits stronger nonlinearity and a wider output range near zero, enhancing the generator's expressive power.

- (ii) The ReLU function is adopted for the remaining layers of the generator to prevent gradient vanishing in deep networks.
 - (iii) The Leaky-ReLU function is used in other discriminator layers to address the issue of dead neurons associated with ReLU, thus improving network stability and generalization.
- (C) **Optimizer:** The Adam optimizer is selected for iterative parameter updates, with the discriminator's learning rate set to 0.0004, four times higher than the generator's.

The detailed training procedure for the WCGAN-GP model is summarized in Algorithm 1.

Algorithm 1 WCGAN-GP training for healthcare fraud detection

Require: Training data: features $X \in \mathbb{R}^{n \times d}$, labels $Y \in \{0, 1\}$; Hyperparameters: $\lambda = 10$ (gradient penalty coefficient), $\alpha = 1.0$ (classification weight), $n_{\text{critic}} = 5$ (critic iterations per generator update), batch size $m = 64$, learning rates $lr_D = 0.0004$, $lr_G = 0.0001$, Adam parameters $\beta_1 = 0.5$, $\beta_2 = 0.999$

Ensure: Trained generator G for synthetic minority class sample generation

Initialize: Generator G_θ and critic D_ϕ with random weights

while training loss has not converged **do**

for $j = 1$ to n_{critic} **do**

 ▷ Update critic n_{critic} times

 Sample mini-batch: $\{X_{\text{real}}, Y_{\text{real}}\}$ from training data

 Sample noise vectors: $Z \sim \mathcal{N}(0, I)$ of size $m \times 100$

 Generate synthetic samples: $X_{\text{fake}} = G(Z, Y_{\text{real}})$

Calculate Wasserstein distance:

$$L_{\text{WD}} = \mathbb{E}[D(X_{\text{real}}, Y_{\text{real}})] - \mathbb{E}[D(X_{\text{fake}}, Y_{\text{real}})]$$

Calculate gradient penalty:

 Sample $\varepsilon \sim \text{Uniform}(0, 1)$

 Interpolate: $\hat{X} = \varepsilon X_{\text{real}} + (1 - \varepsilon) X_{\text{fake}}$

$$\text{GP} = \lambda \cdot \mathbb{E}[(\|\nabla_{\hat{X}} D(\hat{X}, Y_{\text{real}})\|_2 - 1)^2]$$

Total critic loss:

$$L_D = -L_{\text{WD}} + \text{GP}$$

 Update critic: $\phi \leftarrow \text{Adam}(\nabla_\phi L_D, \phi, lr_D, \beta_1, \beta_2)$

end for

Update generator:

 Sample new noise and labels: $Z \sim \mathcal{N}(0, I)$, Y_{fake}

 Generate synthetic samples: $X_{\text{fake}} = G(Z, Y_{\text{fake}})$

 Generator loss: $L_G = -\mathbb{E}[D(X_{\text{fake}}, Y_{\text{fake}})]$

 Update generator: $\theta \leftarrow \text{Adam}(\nabla_\theta L_G, \theta, lr_G, \beta_1, \beta_2)$

end while

The discriminator maximizes the loss function by assigning high probabilities to actual data samples $D(x)$ and low probabilities to generated samples $D(G(z))$. A noise vector is fed into the generator during generator training to produce generated samples. The generator updates its parameters based on the discriminator's feedback, with the aim of maximizing $D(G(z))$. Through this adversarial process, the generator learns to produce samples that the discriminator cannot distinguish from real data, achieving the goal of class prediction.

CGANs use auxiliary conditional information and supervised learning to help facilitate the generation of the targeted sample, in contrast to GANs that only produce generation output. The problems caused by having too many positive samples in comparison with negative samples are greatly reduced by this innovative design. In order to effectively design the conditional frequency distribution in the information, the CGAN framework connects its generative and discriminative components. This characteristic guarantees outstanding fidelity and categorical specificity in the synthesis of samples with predetermined attributes or class labels. The proposed loss function is formulated in Equation (4):

$$\min_G \max_D L(D, G) = \mathbb{E}_{x \sim P_{\text{data}}(x)} [\log D(x|y)] + \mathbb{E}_{z \sim P_{\text{data}}(z)} [\log(1 - D(G(z|y)))]. \quad (4)$$

CGAN can be characterized by high training instability, which results in mode crashes. The mode crash occurs when the generator produces limited and repetitive samples, failing to capture the diversity of the real data distribution. This occurrence indicates the problems CGAN may face when learning and generating complex data distributions, as the generator has trouble mining and simulating the real data distribution to the full extent.

To address the instability of CGAN, we introduced the Wasserstein distance and the gradient penalty term to stabilize the training process and improve sample quality. The Wasserstein distance measures the difference between real and synthetic data distributions, providing a more stable training signal. The incorporation of CGAN and the Wasserstein distance and gradient penalty term combines the advantages of Wasserstein CGAN and gradient penalties with a CGAN, resulting in higher training stability and improved sample quality.

The study [52] introduced Wasserstein GAN and gradient penalty, while WCGAN-GP integrates the conditional information of CGAN with the Wasserstein distance and the gradient penalties. Conditional information y , typically represented as a one-hot encoded class label or attribute vector, guides the generator in producing samples with specific characteristics. The Wasserstein distance provides a stable training signal, while the gradient penalties enforce the Lipschitz constraint, preventing the discriminator from becoming too sensitive. As a result, WCGAN-GP retains the ability to guide generation and mitigates issues like gradient vanishing and training instability, promoting better convergence. We introduced the Wasserstein distance and the gradient penalty term to stabilize the training process and improve sample quality. The Wasserstein distance measures the difference between real and synthetic data distributions, providing a more stable training signal. WCGANs-GP combines the advantages of Wasserstein CGANs with gradient penalties, resulting in higher training stability and improved sample quality.

The definition of the ablation condition was then completed and the ablated model was used for multiple experiments for training and evaluation.

4.3. Evolutionary Feature Selection

Selection and refinement of the input variables are crucial in determining the predictive accuracy and computational efficiency of both machine learning and deep learning frameworks. The inclusion of extraneous or extensive feature collections not only imposes unnecessary computational burdens, but also potentially degrades model performance. In order to extract and preserve a collection of impartial factors that have a significant statistical connection with the goal measure, dimensionality reduction techniques are used. These optimized feature subsets later form the foundation for model development. Interdependent variables, which provide useless information to the method, introduce noise, increase overfitting tendencies, and produce less accurate predictive models. Eliminating features with negligible explanatory power yields multiple benefits, encompassing improved model transparency, enhanced data representation capabilities, and optimized predictive performance. In addition, this scheme makes it easier to make reduced models with better operational effectiveness.

Various dimensionality reduction techniques have been developed, encompassing Principal Component Analysis, t-distributed Stochastic Neighbor integration, Linear Discriminant Analysis, and Wrapper-based methods. GA demonstrate superior characteristics to Principal Component Analysis and Autoencoder-based approaches in feature selection, particularly in their capacity to process nonlinear relationships, evaluate feature importance, preserve interpretability, and exhibit resilience to noisy data and anomalous observations. GA-based feature selection implements an evolutionary optimization process that system-

atically explores the feature space, terminating when convergence criteria are satisfied or the optimization metric reaches a plateau.

We propose the GA-RF method for feature selection, which combines evolutionary optimization with ensemble classification to identify and preserve the most discriminative features. Unlike filter-based methods that rely on statistical measures or wrapper methods that are computationally prohibitive for high-dimensional data, GA-RF directly optimizes for classification performance while efficiently exploring the combinatorial feature space.

The GA-based feature selection is defined as a binary optimization problem in which a chromosome represents a candidate feature subset. Specifically, a chromosome is encoded as a binary vector $\mathbf{b} = [b_1, b_2, \dots, b_n] \in \{0, 1\}^n$, where $b_i = 1$ indicates that the i -th feature is selected and $b_i = 0$ indicates exclusion. The fitness function is precisely defined as follows:

$$\text{Fitness}(\mathbf{b}) = \alpha \cdot \text{F1}_{\text{RF}}(\mathbf{b}) + \beta \cdot \left(1 - \frac{|\mathbf{b}|}{n}\right) \quad (5)$$

where $\text{F1}_{\text{RF}}(\mathbf{b})$ denotes the F1-score achieved by a Random Forest classifier trained on the feature subset encoded by $\mathbf{b}$, $|\mathbf{b}| = \sum_{i=1}^n b_i$ is the number of selected features, $n = 118$ is the total number of features in the original dataset.

Theoretical justification for the fitness function design:

- F1-score term ($\alpha \cdot \text{F1}_{\text{RF}}(\mathbf{b})$): We use F1-score rather than accuracy because (1) in imbalanced fraud detection datasets, accuracy is misleading—the majority class (legitimate claims) dominates, making a trivial “all legitimate” classifier achieve >94% accuracy and (2) F1-score balances precision (avoiding false positives that accuse legitimate providers) and recall (catching actual fraud), which is essential for healthcare fraud detection where both types of errors have significant costs.
- Sparsity term ($\beta \cdot (1 - |\mathbf{b}|/n)$): Encourages compact feature subsets to improve generalization and interpretability.
- Theoretical motivation: The curse of dimensionality states that as feature count increases, the data becomes increasingly sparse, requiring exponentially more samples to achieve reliable estimates. For our 16:1 imbalanced dataset with limited fraud samples, reducing from 118 to 28 features significantly improves the sample-to-feature ratio.
- Weight selection ($\alpha = 0.9, \beta = 0.1$): We prioritize classification performance while still providing sufficient pressure for feature reduction. This asymmetric weighting reflects the practical reality that detection accuracy is paramount, but model interpretability and computational efficiency are also valuable for real-world deployment.

The GA-RF algorithm proceeds through the following steps: (1) **Initialization**: We generate an initial population of size $P = 50$ chromosomes, with each feature selected randomly with probability 0.5; (2) **Evaluation**: The fitness of each chromosome is calculated using Equation (5) through 5-fold cross-validation on the training set; (3) **Selection**: We use tournament selection with tournament size three to select parent chromosomes; (4) **Crossover**: Uniform crossover is performed with probability $p_c = 0.8$ to produce offspring; (5) **Mutation**: Bit-flip mutation is applied with probability $p_m = 0.01$ per gene; and (6) **Replacement**: The population is replaced with offspring while preserving the top 10% elite individuals. This process runs for a maximum of $G = 100$ generations or until convergence, defined as no improvement in the best fitness for 20 consecutive generations.

The GA-RF method identified a subset of 28 features, achieving 97.05% recall on the test dataset. These results demonstrate the approach’s effectiveness in feature selection, along with strong representativeness and predictive performance. The GA-RF model is particularly valuable for tasks requiring accurate classification and dimensionality reduction. The model can explore sequential effects between features and identify non-intuitive opti-

mal subsets, especially in cases with intricate feature interactions. The GA-RF framework is illustrated in Figure 4.

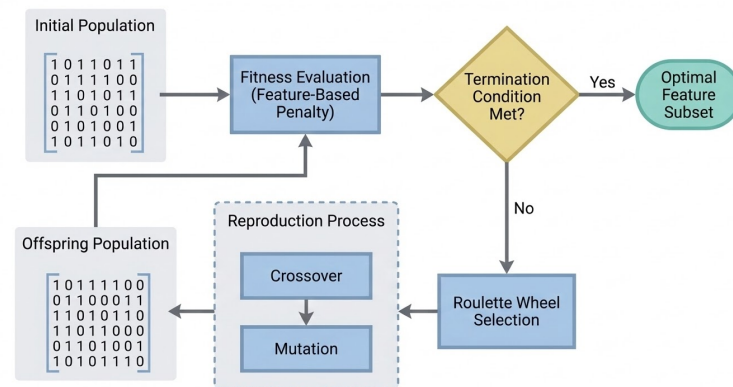


Figure 4. The GA-RF algorithm identifies the most discriminative feature subset through evolutionary search, using Random Forest classification performance as fitness to guide the selection process towards optimal feature reduction.

4.4. Prediction

The healthcare dataset is used to train the HFD model after data sample augmentation and the features extraction. The complex association between input data patterns and corresponding fault classifications is modeled through a transformation mechanism implemented within a neural network architecture. The underlying connections between raw input features and their respective medical labels can be learned using this scheme. The hierarchical construction of the neural network enables the extraction of more abstract features, helping to establish a strong tracking relationship between the input space and the outcomes of fault categorization. The RF algorithms are connected simultaneously after feature extraction. In addition to providing in-depth analysis of the proposed methodology's performance, the following sections provide in-depth analysis of the proposed methodology's effectiveness and an important discussion of model performance metrics.

5. Experiments

5.1. Dataset Description and Verification Metrics

This experiment uses a healthcare dataset sourced from Neusoft, a recognized technology firm that specializes in health insurance systems. The dataset can be accessed online at <http://www.fwwb.org.cn/news/show/462> (accessed on 3 May 2024). For the University Student Innovation and Entrepreneurship Competition 2024, it was created from scratch. It contains 16,000 instances, each characterized by 118 attributes, and categorizes genuine medical claims into four categories: care histories, spending behaviors, cost structures, and forgery annotations, as shown in Table 2.

Past treatment samples are used mainly to obtain clinic treatment records and monthly clinic visits. Consumption measures track total monthly expenditures for different groups, including medication purchases, medical tests, accommodation fees, and subsidy reimbursements. To capture spending habits, cost ratios measure the share of care in general expenses, the proportion of total usage allocated to medicines, and the percentage of patient out-of-pocket payments. These components provide a comprehensive and reliable basis for studying different aspects of medical utilization and payment behavior.

Table 2. The detailed information of the dataset.

Category	Feature	Description
Monthly healthcare treatment	Monthly visits	Number of days visiting hospitals in one month
		Number of months of visit
	Monthly number of visits	Monthly number of healthcare hospitals MAX
		Monthly number of visiting hospitals AVG
		Monthly clinic visits MAX
	Number of months of treatment	Monthly clinic visits AVG
		Number of clinic visits to SUM
		Hospital visit days MAX
		Hospital visit days AVG
		Number of months of visit
The amount of consumption	Monthly overall amount	Monthly pooling amount MAX
		Monthly pooling amount AVG
		Hospital Coordinating gold MAX
	Drug amount	Hospital Coordinating gold AVG
		Hospital drug product AVG
		Hospital drug product MAX
		Hospital drug product AVG
		Amount of drug fee incurred SUM
		Drug cost amount SUM
		Declaration amount of drug fee SUM
		Amount of valuable drugs occurred SUM
		Amount of Chinese patent medicine fee incurred SUM
		Amount of Chinese herbal medicine fee incurred SUM
	The cost of the inspection required	Amount of inspection fee incurred SUM
		Amount of the inspection fee SUM
		Declaration amount of the inspection fee SUM
		Precious inspection fee amount SUM
		Amount of treatment cost incurred SUM
		Treatment cost amount SUM
		Declaration amount of treatment fee SUM
		Amount of operation incurred SUM
		Self-cost amount SUM
		Declaration amount of surgery fee SUM
	Hospitality	Amount of bed fee incurred SUM
		Declaration amount of bed fee SUM
		Amount of healthcare materials occurred SUM
		Amount of high-priced material occurrence SUM
		Amount of healthcare materials cost SUM
		Composition transfusion declared amount SUM
		Other amount occurred SUM
	Subsidy declaration	Other declared amounts SUM
		Declaration amount of one-time healthcare materials SUM
		Standard amount of the starting payment line MAX
		Self-proportion proportion above the minimum payment standard SUM
		Individual healthcare assistance SUM
		Max. amount above SUM
		Amount paid by the basic pooling fund SUM

Table 2. *Cont.*

Category	Feature	Description
		Amount paid by Medicaid fund SUM
		Amount of urban and rural assistance subsidies SUM
		Basic personal account payment for SUM
		Non-account payment amount SUM
		This approved amount is SUM
		Subsidy approval amount SUM
		healthcare assistance hospital application for SUM
		Disabled military allowance SUM
		Civil Affairs aid subsidy SUM
Cost ratio		Urban and rural entitled treatment subsidies SUM
		The proportion of drugs in the total amount
		The proportion of drugs paid for by individuals
		The total cost of the inspection is proportional to the total amount
		The proportion of individual payment of inspection fees
		The proportion of treatment costs in the total amount
		The proportion of the treatment costs paid by individuals

The dataset is divided into two categories: legitimate samples (denoted as +) and fraudulent samples (denoted as −). The occurrence rates for legitimate samples are 94.1% (15,056), while fraudulent samples have an occurrence rate of 5.9% (944). Thus, a binary classification task with an imbalance ratio of 16:1 is defined as the fraud forecast problem. According to this ratio, there are approximately 16 legitimate examples for every fraudulent record.

To ensure rigorous and reproducible evaluation, we employ a stratified train-test split with the following protocol: (1) **Stratified sampling**: The dataset is partitioned using stratified sampling to preserve the class distribution ratio in both training and test sets, ensuring that the minority class is adequately represented in the test set; (2) **Random seed**: We set a fixed random seed ($seed = 42$) for reproducibility; (3) **Data leakage prevention**: The test set is held out completely during model training, hyperparameter tuning, and feature selection. All data augmentation (via WCGAN-GP) and feature selection (via GARF) are performed exclusively on the training set; (4) **Split ratio**: The empirical framework allocates 80% of the dataset for model training and reserves 20% for evaluation purposes; and (5) **Multiple runs**: To assess performance variability, we report results averaged over five independent runs with different random seeds ($seeds \in \{42, 123, 456, 789, 1024\}$), with standard deviations reported in all tables. We note that while k -fold cross-validation is commonly used for model selection, we employ hold-out validation in this study because (a) the dataset size ($n = 16,000$) provides sufficient test samples for reliable evaluation; (b) the severe class imbalance (16:1) makes stratified hold-out more practical than k -fold for ensuring minority class representation in each fold; and (c) the computational cost of training WCGAN-GP makes repeated k -fold training prohibitive.

We performed preprocessing activities to ensure correct data analysis and preparation for further analysis. The dataset was optimized to identify significant factors in detecting fraud. Statistics are carefully structured and measured by category, following the classification of the data into training and testing subsets.

5.2. WCGAN-GP Evaluation

The experimental data were divided into two groups, that is, training and testing, using machine learning protocols, which used a split percentage of 8:2. The imbalance in distribution among classes was addressed using four different oversampling methods that include ROS, SMOTE, ADASYN, and our WCGAN-GP. The WCGAN-GP approach,

in particular, was specifically made to generate more high-quality data examples using Wasserstein distance measures and gradient penalty considerations.

Figure 5 shows the evolution of the loss during WCGAN-GP training, exhibiting a period of initial divergence followed by stable convergence after approximately 10,000 iterations. To quantitatively assess the training dynamics, we analyzed several key metrics: (1) **Convergence speed**: Both generator and discriminator losses stabilized within 10,000 iterations, demonstrating efficient training; (2) **Loss stability**: After convergence, the generator loss maintained a low variance of 0.15 ± 0.02 , while the discriminator loss stabilized at 0.85 ± 0.03 ; and (3) **Loss ratio**: The generator-to-discriminator loss ratio converged to 0.18 ± 0.01 , indicating a balanced adversarial dynamic. These quantitative indicators confirm that the model achieved a stable equilibrium, with neither component dominating the training process.

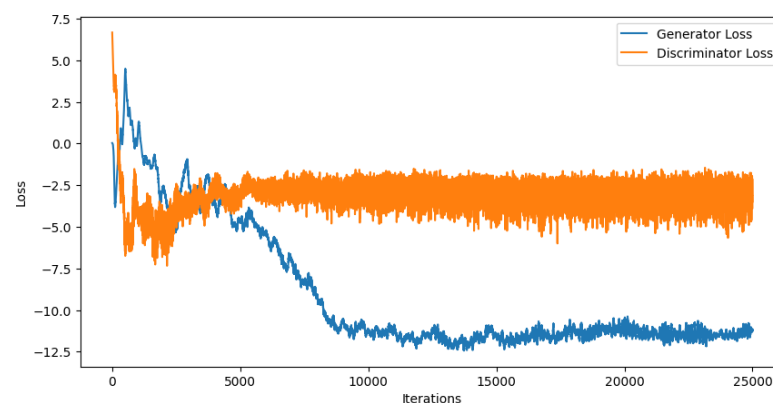


Figure 5. WCGAN-GP training dynamics showing generator and discriminator loss convergence over 10,000 iterations. The x-axis uses a scale of 2000 iterations per major tick for enhanced visualization of training progression. The model achieves stable equilibrium after the initial training phase, with losses plateauing at $G_{loss} \approx 0.15$ and $D_{loss} \approx 0.85$.

The convergence to a generator loss of 0.15 and discriminator loss of 0.85 after 10,000 iterations, as shown in Figure 5, demonstrates successful model training. A trained GAN is considered successful when the generator loss plateaus below 0.2 and the discriminator loss plateaus above 0.8, as these values represent a balanced adversarial interaction where neither component dominates. This equilibrium enables the GAN to produce highly realistic synthetic samples while maintaining robust discrimination capabilities.

Table 3 compares the performance metrics (precision, recall, and F1-score) of four oversampling methods: ROS, SMOTE, ADASYN, and our WCGAN-GP, on the test set. The metrics were calculated separately for each class, and a weighted average, based on the class distribution, was used to assess overall model performance. All results are reported as mean $\pm$ standard deviation over five independent runs. As shown in Table 3, the WCGAN-GP method achieved the highest weighted precision of $96.82 \pm 0.4\%$, weighted recall of $96.94 \pm 0.3\%$, and weighted F1-score of $96.51 \pm 0.3\%$. Notably, WCGAN-GP demonstrates superior performance on the minority class (fraudulent claims) with a precision of $88.62 \pm 1.2\%$ and F1-score of $52.61 \pm 0.9\%$, significantly outperforming traditional oversampling methods. In contrast, ADASYN achieved a weighted F1-score of $95.08 \pm 0.5\%$, while SMOTE performed lower with $94.97 \pm 0.6\%$. The comparative analysis demonstrates that the adversarial generative network method significantly outperforms SMOTE and ADASYN in terms of both precision and robustness. These findings demonstrate that WCGAN-based approaches are highly effective for tasks requiring high precision on imbalanced data, such as fraud detection.

Our findings indicate that the WCGAN-GP model serves as an effective tool to detect fraud in imbalanced high-dimensional datasets.

Table 3. Comparison of different oversampling methods (mean $\pm$ std over five runs). The table reports precision, recall, and F1-score for (1) majority class (legitimate claims), (2) minority class (fraudulent claims), and (3) weighted average based on class distribution. The bold values indicate the best performance in each category.

	Method	Precision	Recall	F1-Score
Majority class	ROS	96.98 $\pm$ 0.3%	99.37 $\pm$ 0.2%	98.17 $\pm$ 0.2%
	SMOTE	97.53 $\pm$ 0.4%	97.05 $\pm$ 0.5%	97.29 $\pm$ 0.4%
	ADASYN	97.59 $\pm$ 0.4%	97.11 $\pm$ 0.5%	97.35 $\pm$ 0.4%
	WCGAN-GP	96.97 $\pm$ 0.3%	99.79 $\pm$ 0.2%	98.40 $\pm$ 0.2%
Minority class	ROS	75.31 $\pm$ 2.1%	38.16 $\pm$ 1.8%	50.65 $\pm$ 1.5%
	SMOTE	46.11 $\pm$ 2.3%	50.66 $\pm$ 2.0%	48.28 $\pm$ 1.8%
	ADASYN	47.31 $\pm$ 2.2%	51.97 $\pm$ 1.9%	49.53 $\pm$ 1.7%
	WCGAN-GP	88.62 $\pm$ 1.2%	37.64 $\pm$ 1.5%	52.61 $\pm$ 0.9%
Weighted	ROS	95.96 $\pm$ 0.5%	96.47 $\pm$ 0.4%	95.91 $\pm$ 0.4%
	SMOTE	95.09 $\pm$ 0.7%	94.84 $\pm$ 0.6%	94.97 $\pm$ 0.6%
	ADASYN	95.21 $\pm$ 0.6%	94.97 $\pm$ 0.5%	95.08 $\pm$ 0.5%
	WCGAN-GP	96.82 $\pm$ 0.4%	96.94 $\pm$ 0.3%	96.51 $\pm$ 0.3%

Note: WCGAN-GA-RF is our proposed scheme.

The superior performance of WCGAN-GP over traditional oversampling methods can be attributed to fundamental differences in synthetic sample generation. SMOTE and ADASYN rely on linear interpolation between minority class samples, which limits their ability to capture complex, nonlinear relationships in high-dimensional feature spaces (118 features in our dataset). This approach becomes increasingly problematic in severely imbalanced scenarios (16:1 ratio) where the minority class occupies a sparse region of the feature space. Linear interpolation may generate synthetic samples that fall outside the true data distribution, introducing noise and reducing model performance. In contrast, WCGAN-GP learns the underlying data distribution through adversarial training, enabling it to generate diverse and realistic samples that better represent the complexity of fraudulent claims. The Wasserstein distance provides stable gradients even when the real and generated distributions have disjointed supports, while the gradient penalty ensures the discriminator satisfies the Lipschitz constraint, leading to more stable training dynamics. This theoretical foundation enables WCGAN-GP to produce high-quality synthetic samples that effectively augment the minority class without introducing artifacts.

A notable observation from Table 3 is the precision–recall trade-off for the minority class: WCGAN-GP achieves the highest precision (88.62 $\pm$ 1.2%) but a relatively lower recall (37.64 $\pm$ 1.5%) compared to ADASYN (51.97 $\pm$ 1.9%). This trade-off arises because WCGAN-GP generates high-quality synthetic samples that lead to a more conservative classifier—while it identifies fraud cases with high confidence (evidenced by the high precision), it may miss some genuine fraud cases. Despite the lower recall, WCGAN-GP still achieves the best F1-score (52.61 $\pm$ 0.9%) among all methods, indicating an overall better balance. Furthermore, in healthcare fraud detection, high precision is often more desirable than high recall because false positives (incorrectly flagging legitimate claims as fraudulent) can erode trust and cause unnecessary investigations. The 88.62% precision means that 88.62% of claims flagged as fraudulent are truly fraudulent, reducing the burden of manual review on legitimate claims.

5.3. GA-RF Evaluation

Following the implementation of GA as our feature selection methodology, we identified 28 statistically independent features of the dataset. The comparison of feature importance before and after WCGAN-GP augmentation is presented in Figures 6 and 7. Notably, after data augmentation, the importance distribution becomes more balanced across features, with several features showing increased discriminative power for fraud detection. The augmented data enables the GA-RF algorithm to identify more robust feature subsets that better capture the characteristics of fraudulent claims. Our WCGAN-GA-RF framework achieves outstanding performance metrics, with a recall rate of 97.05% and a precision of 96.47%, with only 35% of the original feature set (28 out of 80 features) remaining for fraud detection in high-dimensional and severely imbalanced datasets.

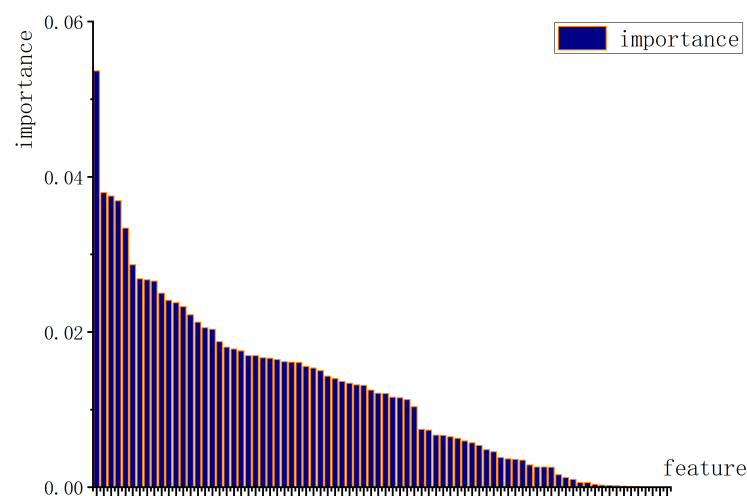


Figure 6. Feature importance analysis on the original imbalanced dataset reveals that Random Forest assigns higher importance to features distinguishing the majority class (legitimate claims), while fraud-discriminative features receive lower scores due to severe class imbalance (16:1 ratio).

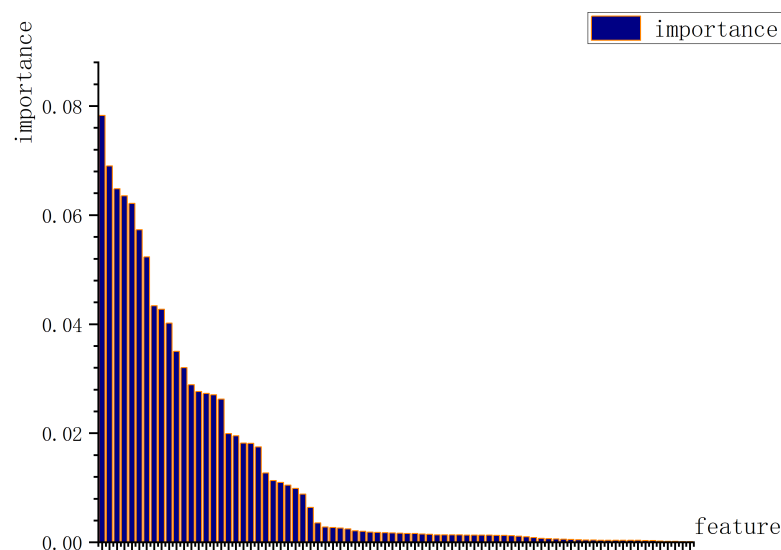


Figure 7. Feature importance analysis on WCGAN-GP-augmented balanced data shows that Random Forest now identifies different features as important, enabling more discriminative feature subset selection that captures genuine fraud patterns.

Despite a 35% reduction in features, the proposed model significantly outperforms conventional sampling methods (see Figure 8), including ROS, SMOTE, and ADASYN.

The WCGAN-GP architecture infrastructure incorporates the Wasserstein distance and gradient penalty systems, which properly address class imbalance issues. The GA approach has several advantages over its competitors (1) improved ability to explore nonlinear feature relationships, (2) more flexibility in exploration of feature space, (3) improved interpretability and integration of human insight, (4) efficient handling of high-dimensional data structures, and (5) robust performance in the presence of noise and anomalies. A comparison of these five models is presented in Table 4.

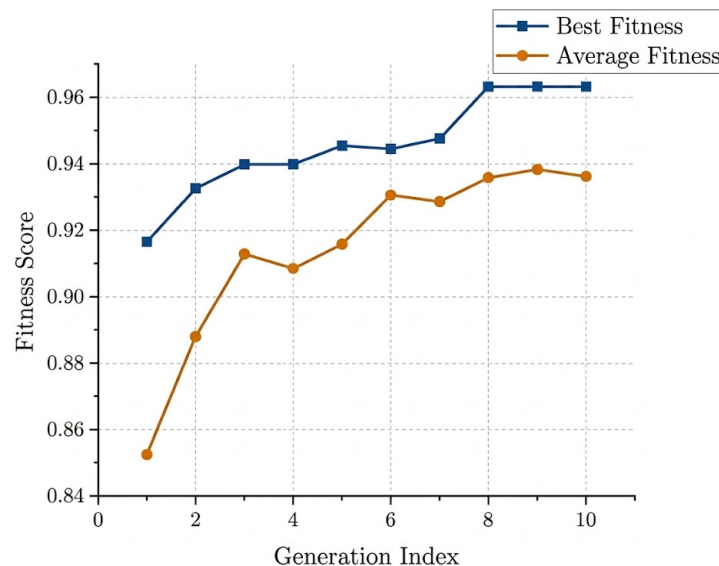


Figure 8. The performance of the genetic algorithm.

Table 4. Comparison of precision of different models on HFD. All models use Random Forest classifier. The key difference is the oversampling method and feature selection strategy. WCGAN-GA-RF uses only 28 features (35% of the original 80), achieving superior performance with reduced dimensionality.

	Features	Precision	Recall	F1-Score
ROS	80	95.96%	96.47%	95.91%
SMOTE	80	95.09%	94.84%	94.97%
ADASYN	80	95.21%	94.97%	95.08%
WCGAN-GP	80	96.82%	96.94%	96.51%
WCGAN-GA-RF	28	96.47%	97.05%	96.26%

Note: **WCGAN-GA-RF** is our proposed scheme.

5.4. Ablation Study

To evaluate the contribution of each component in the proposed WCGAN-GA-RF framework, we conduct a comprehensive ablation study. The results, shown in Table 5, demonstrate that the synergistic combination of WCGAN-GP data augmentation, GA-based feature selection, and RF classification yields superior performance compared to individual components.

Key findings from the ablation study are the following: (1) **WCGAN-GP contribution:** Comparing rows 1–2 shows that WCGAN-GP data augmentation improves F1-score from 94.95% to 96.51% (+1.56%), demonstrating the effectiveness of generative augmentation for handling class imbalance. (2) **GA-RF contribution:** Comparing rows 2 and 5 shows that GA-RF feature selection reduces feature count from 80 to 28 (65% reduction) while maintaining competitive F1-score (96.51% vs. 96.26%, only 0.25% drop), demonstrating efficient dimensionality reduction. (3) **Synergistic effect:** The full WCGAN-GA-RF achieves

the best balance of precision and recall with only 35% of the original features, outperforming individual components in terms of efficiency–accuracy trade-off.

Table 5. Ablation study and contribution of each component (mean $\pm$ std over five runs).

Model Variant	Feat.	Precision	Recall	F1
Baseline (no augmentation)	80	95.32 $\pm$ 0.7%	94.87 $\pm$ 0.6%	94.95 $\pm$ 0.6%
WCGAN-GP only	80	96.82 $\pm$ 0.4%	96.94 $\pm$ 0.3%	96.51 $\pm$ 0.3%
GA-RF only (with SMOTE)	28	95.78 $\pm$ 0.6%	95.92 $\pm$ 0.5%	95.85 $\pm$ 0.5%
WCGAN + RF (full features)	80	96.91 $\pm$ 0.3%	97.03 $\pm$ 0.3%	96.97 $\pm$ 0.3%
Full WCGAN-GA-RF	28	96.47 $\pm$ 0.5%	97.05 $\pm$ 0.4%	96.26 $\pm$ 0.4%

Note: **Full WCGAN-GA-RF** is our proposed scheme.

The computational costs of the proposed framework are as follows: (1) **WCGAN-GP training**: $O(n \cdot d \cdot I)$ where $n = 12,800$ training samples, $d = 80$ features, and $I = 10,000$ iterations. On our hardware (NVIDIA RTX 3080 GPU), WCGAN-GP training takes approximately 15 min. (2) **GA-RF feature selection**: $O(P \cdot G \cdot k \cdot \text{RF}_{\text{train}})$ where $P = 50$ population size, $G = 100$ generations, and k -fold cross-validation ($k = 5$). This takes approximately 8 min on a standard CPU. (3) **RF classification**: Training takes $O(n \cdot d_{\text{selected}} \cdot t \cdot \log t)$ where $d_{\text{selected}} = 28$ features and $t = 100$ trees, requiring less than 1 min. The total training pipeline completes in under 30 min, making it practical for periodic model updates in real-world deployment scenarios.

6. Conclusions

Due to its complex systems and huge financial moves, the healthcare industry has emerged as a premier destination for fraud activities. The changing demography, including an aging population, offers these insurance programs even more money, adding even more stress to their balance. Fraud eventually threatens both the economic stability and the healthcare of the individuals who rely on these services. Advances in machine learning, improved data collection techniques, and the expansion of public state data all contribute to the development of new approaches to fight back. With these tools, fraud can be identified more complex and powerfully through advanced analyses. It has become so important to develop powerful computerized systems that identify suspicious trends and contact related functions right away.

Severe class imbalance, high-dimensional feature spaces, and multiple-class distributions are common hallmarks of HFD datasets. Conventional HFD models trained on such imbalanced datasets generally exhibit exceptional classification accuracy for majority-class samples but underperform in minority-class instances. In addition, the feature selection process poses a major problem because it affects both the accuracy of the monitoring and the computational efficiency. We present a novel framework that combines WCGAN-GP with GA features optimization and prediction modules to target these limitations. The WCGAN-GP component helps to oversample the minority samples in an efficient manner, and the procedure of the GA-based feature selection process meticulously removes the redundant features, and a closing classification structure is pursued. Experimental findings indicate that the proposed design drastically beats current techniques in correcting class imbalance problems in healthcare claim information. Our feature selection strategy, which is based on GA, has superior measures of efficiency. The proposed HFD system is effective in detecting fraud within the system due to the built-in architecture.

Future research directions refer to various interesting directions that can be taken to improve the proposed framework further. To begin with, the development of deep learning classifiers is a natural continuation of this project. In particular, we will explore (1) **transformer-based architectures** to discover long-range dependency in sequential

healthcare claim data, which could enhance the ability to detect complex fraud patterns across the dimension of time; (2) **Graph Neural Networks (GNNs)** to discover relationships among healthcare providers, patients, and claims, which could be used to detect network-based fraud schemes; and (3) **hybrid deep learning models** to combine the representational power of neural networks and the interpretability of tree-based models. These deep learning methods may substitute the existing Random Forest classifier in the WCGAN-GA-RF structure and would possibly enhance detection accuracy without reducing interpretability in attention mechanisms and feature attribution techniques.

Second, the domain of advanced optimization methods has great potential in enhancing the computational effectiveness and the performance of the model. We suggest investigating (1) **Bayesian optimization** for hyperparameter tuning of WCGAN-GP to replace the current grid search approach, which, as a probabilistic optimization method, can reduce training time while identifying optimal configurations; (2) **multi-objective evolutionary algorithms** (e.g., NSGA-II, MOEA/D) for feature selection, enabling simultaneous optimization of multiple criteria such as classification accuracy, feature sparsity, and computational cost; and (3) **adaptive learning rate schedulers** and **neural architecture search** algorithms to automatically optimize the network structure of the generator and discriminator. These optimization measures may minimize the computational overhead of the framework while maintaining or improving detection performance.

Third, future work will be concerned with practical deployment considerations. This involves the creation of explainable AI (XAI) techniques to obtain interpretable fraud detection results for regulatory compliance and auditor review, implementing real-time streaming architectures to generate low-latency results for incoming claims, and exploring federated learning methods to train models across various healthcare organizations while maintaining data privacy. Additionally, integrating multi-modal data sources (e.g., medical images, clinical notes, temporal claim sequences) could further enhance the model's ability to detect advanced fraud schemes that would evade detection when working with claims data alone.

Despite the promising results, this study has several limitations that warrant acknowledgement. First, the empirical validation is conducted on a single healthcare claims dataset from one region (China), and generalization to other healthcare systems with different fraud patterns and regulatory environments requires further investigation. Second, the framework assumes that labeled fraud data is available for training, which may not hold in deployment scenarios where fraud labels are scarce or unreliable. Third, while our approach achieves high detection accuracy, the real-world economic impact and return-on-investment of deploying this system remain to be validated through operational pilots. Finally, regulatory and privacy considerations, particularly regarding the use of synthetic patient data and cross-institutional model deployment, are important practical concerns that are beyond the scope of this methodological study but essential for eventual adoption.

Author Contributions: J.C. and S.W. led the writing of the main text of the manuscript. The methodology and conceptualization were developed by J.C. and S.W.; J.S., Y.T., J.C. and Y.Z. were responsible for the preparation of the figures. All authors have read and agreed to the published version of the manuscript.

Funding: This work is supported in part by the Zhejiang University of Science and Technology integrated industry education project under grant No.: KJ-2024-270 and by the National College Student Innovation and Entrepreneurship Training Program under grant No.: 202511057060.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The healthcare claims dataset used in this study was provided by Neusoft Corporation under a data use agreement for academic research. Due to the proprietary nature of the data containing sensitive patient information, the dataset cannot be publicly released. However, the dataset statistics and preprocessing details are provided in Section 5.1 to enable reproducibility assessment. The source code for the WCGAN-GA-RF framework is available from the corresponding author upon reasonable request, subject to appropriate data use agreements.

Acknowledgments: The authors sincerely thank the reviewers and editors for their constructive comments and valuable suggestions, which have significantly improved the quality of this manuscript.

Conflicts of Interest: The authors declare no conflict of interest.

References

1. du Preez, A.; Bhattacharya, S.; Beling, P.; Bowen, E. Fraud detection in healthcare claims using machine learning: A systematic review. *Artif. Intell. Med.* **2024**, *160*, 103061. [CrossRef]
2. National Healthcare Security Administration. 2020 Statistical Report on the Advancement of the Medical Security Sector. Available online: https://www.nhsa.gov.cn/art/2021/3/8/art_7_4590.html (accessed on 7 January 2025).
3. National Healthcare Security Administration. Statistical Report on the Development of Medical Security in 2020 Health Expenditure Fact Sheet. Available online: https://www.nhsa.gov.cn/art/2021/3/8/art_7_4590.html?utm_source=chatgpt.com (accessed on 7 January 2025).
4. Chandola, V.; Banerjee, A.; Kumar, V. Anomaly detection: A survey. *ACM Comput. Surv.* **2009**, *41*, 1–58. [CrossRef]
5. He, H.; Garcia, E.A. Learning from imbalanced data. *IEEE Trans. Knowl. Data Eng.* **2009**, *21*, 1263–1284. [CrossRef]
6. Guyon, I.; Elisseeff, A. An introduction to variable and feature selection. *J. Mach. Learn. Res.* **2003**, *3*, 1157–1182.
7. Beygelzimer, A.; Dasgupta, S.; Langford, J. Importance weighted active learning. In *Proceedings of the 26th Annual International Conference on Machine Learning*; Association for Computing Machinery: New York, NY, USA, 2009; pp. 49–56.
8. Tubishat, M.; Tbaishat, D.; Al-Zoubi, A.M.; Hraiz, A.E.; Habib, M. Leveraging evolutionary algorithms with a dynamic weighted search space approach for fraud detection in healthcare insurance claims. *Knowl.-Based Syst.* **2025**, *317*, 113436. [CrossRef]
9. Matloob, I.; Khan, S.A.; Rukaiya, R.; Alfraihi, H.; Khan, J.A. Healthcare fraud detection using adaptive learning and deep learning techniques. *Evol. Syst.* **2025**, *16*, 72. [CrossRef]
10. Mao, Y.; Li, Y.; Xu, B.; Han, J. XGAN: A Medical Insurance fraud Detector based on GAN with XGBoost. *J. Inf. Hiding Multimed. Signal Process.* **2024**, *15*, 36–52.
11. Dong, Y.; Yao, J.; Wang, J.; Liang, Y.; Liao, S.; Xiao, M. Dynamic fraud detection: Integrating reinforcement learning into graph neural networks. In *2024 6th International Conference on Data-Driven Optimization of Complex Systems (DOCS)*; IEEE: Hangzhou, China, 2024; pp. 818–823.
12. Nugraha, R.A.; Pardede, H.F.; Subekti, A. Oversampling based on generative adversarial networks to overcome imbalance data in predicting fraud insurance claim. *Kuwait J. Sci.* **2022**, *49*, 1–12. [CrossRef]
13. Yang, W.S.; Hwang, S.Y. A process-mining framework for the detection of healthcare fraud and abuse. *Expert Syst. Appl.* **2006**, *31*, 56–68. [CrossRef]
14. Zhang, C.; Xiao, X.; Wu, C. Medical fraud and abuse detection system based on machine learning. *Int. J. Environ. Res. Public Health* **2020**, *17*, 7265. [CrossRef]
15. Massi, M.C.; Ieva, F.; Lettieri, E. Data mining application to healthcare fraud detection: A two-step unsupervised clustering method for outlier detection with administrative databases. *BMC Med. Inform. Decis. Mak.* **2020**, *20*, 160. [CrossRef]
16. Kanksha; Bhaskar, A.; Pande, S.; Malik, R.; Khamparia, A. An intelligent unsupervised technique for fraud detection in healthcare systems. *Intell. Decis. Technol.* **2021**, *15*, 127–139.
17. Chen, R.; Zhang, H.; Lin, K. A graph-based method for health care joint fraud detection. In *Proceedings of the 2020 9th International Conference on Computing and Pattern Recognition*; Association for Computing Machinery: New York, NY, USA, 2020; pp. 122–129.
18. Lin, C.; Lin, C.M.; Li, S.T.; Kuo, S.C. Intelligent physician segmentation and management based on KDD approach. *Expert Syst. Appl.* **2008**, *34*, 1963–1973. [CrossRef]
19. Snorovichina, V.; Zaytsev, A. Unsupervised anomaly detection for discrete sequence healthcare data. In *International Conference on Analysis of Images, Social Networks and Texts*; Springer International Publishing: Cham, Switzerland, 2020; pp. 391–403.
20. Sun, C.; Li, Q.; Li, H.; Shi, Y.; Zhang, S.; Guo, W. Patient cluster divergence based healthcare insurance fraudster detection. *IEEE Access* **2018**, *7*, 14162–14170. [CrossRef]
21. Lu, J.; Lin, K.; Chen, R.; Lin, M.; Chen, X.; Lu, P. Health insurance fraud detection by using an attributed heterogeneous information network with a hierarchical attention mechanism. *BMC Med. Inform. Decis. Mak.* **2023**, *23*, 62. [CrossRef]

22. Kennedy, R.K.; Salekshahrezaee, Z.; Villanustre, F.; Khoshgoftaar, T.M. Iterative cleaning and learning of big highly-imbalanced fraud data using unsupervised learning. *J. Big Data* **2023**, *10*, 106. [CrossRef]
23. Suesserman, M.; Gorny, S.; Lasaga, D.; Helms, J.; Olson, D.; Bowen, E.; Bhattacharya, S. Procedure code overutilization detection from healthcare claims using unsupervised deep learning methods. *BMC Med. Inform. Decis. Mak.* **2023**, *23*, 196. [CrossRef]
24. Taha, A.; Cosgrave, B.; McKeever, S. Using feature selection with machine learning for generation of insurance insights. *Appl. Sci.* **2022**, *12*, 3209. [CrossRef]
25. Ekin, T.; Frigau, L.; Conversano, C. Health care fraud classifiers in practice. *Appl. Stoch. Model. Bus. Ind.* **2021**, *37*, 1182–1199. [CrossRef]
26. He, H.; Hawkins, S.; Graco, W.; Yao, X. Application of genetic algorithm and k-nearest neighbour method in real world medical fraud detection problem. *J. Adv. Comput. Intell. Inform.* **2000**, *4*, 130–137. [CrossRef]
27. Bauder, R.A.; Khoshgoftaar, T.M. The effects of varying class distribution on learner behavior for medicare fraud detection with imbalanced big data. *Health Inf. Sci. Syst.* **2018**, *6*, 9. [CrossRef]
28. Chen, J.P.; Lu, P.; Yang, F.; Chen, R.; Lin, K. Medical insurance fraud detection using graph neural networks with spatio-temporal constraints. *J. Netw. Intell.* **2022**, *7*, 480–498.
29. Fursov, I.; Kovtun, E.; Rivera-Castro, R.; Zaytsev, A.; Khasyanov, R.; Spindler, M.; Burnaev, E. Sequence embeddings help detect insurance fraud. *IEEE Access* **2022**, *10*, 32060–32074. [CrossRef]
30. Matschak, T.; Prinz, C.; Rampold, F.; Trang, S. Show Me Your Claims and I'll Tell You Your Offenses: Machine Learning-Based Decision Support for Fraud Detection on Medical Claim Data. In Proceedings of the Hawaii International Conference on System Sciences 2022 (HICSS-55), Virtual, 4–7 January 2022.
31. Zhang, G.; Fu, S.; Xu, X.; Qi, L.; Zhang, X.; Dou, W. An anti-fraud framework for medical insurance based on deep learning. In *Advanced Data Mining and Applications ADMA 2019*; Lecture Notes in Computer Science; Springer: Cham, Switzerland, 2019; Volume 11888. [CrossRef]
32. Mayaki, M.Z.A.; Riveill, M. Multiple inputs neural networks for fraud detection. In *International Conference on Machine Learning, Control, and Robotics (MLCR)*; IEEE: Piscataway, NJ, USA, 2022; pp. 8–13.
33. Kumari, A.; Pun, N.S.; Sonbhadra, S.K.; Agarwal, S. Impact of the composition of feature extraction and class sampling in medicare fraud detection. In *International Conference on Neural Information Processing*; Springer International Publishing: Cham, Switzerland, 2022; pp. 639–658.
34. Bounab, R.; Zarour, K.; Guelib, B.; Khelifa, N. Enhancing Medicare Fraud Detection Through Machine Learning: Addressing Class Imbalance with SMOTE-ENN. *IEEE Access* **2024**, *12*, 54382–54396. [CrossRef]
35. Kumaraswamy, N.; Ekin, T.; Park, C.; Markey, M.K.; Barner, J.C.; Rascati, K. Using a Bayesian Belief Network to detect healthcare fraud. *Expert Syst. Appl.* **2024**, *238*, 122241. [CrossRef]
36. Mohanta, A.; Panigrahi, S. Health Insurance Fraud Detection Using Feature Selection and Ensemble Machine Learning Techniques. In *Advances in Distributed Computing and Machine Learning: Proceedings of ICADCML 2023*; Springer Nature: Singapore, 2023; pp. 197–207.
37. Sun, C.; Cui, H.; Zhou, W.; Nie, W.; Wang, X.; Yuan, Q. Epileptic seizure detection with EEG textural features and imbalanced classification based on EasyEnsemble learning. *Int. J. Neural Syst.* **2019**, *29*, 1950021. [CrossRef]
38. Liu, X.Y.; Wu, J.; Zhou, Z.H. Exploratory undersampling for class-imbalance learning. *IEEE Trans. Syst. Man Cybern. Part B (Cybern.)* **2008**, *39*, 539–550.
39. Wang, Y.; Yang, L. A robust loss function for classification with imbalanced datasets. *Neurocomputing* **2019**, *331*, 40–49. [CrossRef]
40. Saia, R. Unbalanced Data Classification in Fraud Detection by Introducing a Multidimensional Space Analysis. In *Proceedings of the 3rd International Conference on Internet of Things, Big Data and Security (IoTBDs 2018)*; SciTePress: Setúbal, Portugal, 2018; pp. 29–40. Available online: <https://www.scitepress.org/Papers/2018/66630/66630.pdf> (accessed on 7 January 2025).
41. Yu, H.; Yang, X.; Zheng, S.; Sun, C. Active learning from imbalanced data: A solution of online weighted extreme learning machine. *IEEE Trans. Neural Netw. Learn. Syst.* **2018**, *30*, 1088–1103. [CrossRef]
42. Naidoo, K.; Marivate, V. Unsupervised anomaly detection of healthcare providers using generative adversarial networks. In *Conference on e-Business, e-Services, and e-Society*; Springer: Cham, Switzerland, 2020; pp. 419–430.
43. He, H.; Graco, W.; Yao, X. Application of genetic algorithm and k-nearest neighbour method in medical fraud detection. In *Asia-Pacific Conference on Simulated Evolution and Learning*; Springer: Berlin/Heidelberg, Germany, 1998; pp. 74–81.
44. Majhi, S.K.; Bhattacharya, S.; Pradhan, R.; Biswal, S. Fuzzy clustering using salp swarm algorithm for automobile insurance fraud detection. *J. Intell. Fuzzy Syst.* **2019**, *36*, 2333–2344. [CrossRef]
45. Yan, C.; Li, M.; Liu, W.; Qi, M. Improved adaptive genetic algorithm for the vehicle Insurance Fraud Identification Model based on a BP Neural Network. *Theor. Comput. Sci.* **2020**, *817*, 12–23. [CrossRef]
46. Lingeswari, R.; Brindha, S. Efficient loss updated XGBoost with deep emended genetic algorithm for detecting online fraudulent transactions. *Multimed. Tools Appl.* **2024**, *83*, 84471–84494. [CrossRef]

47. Goodfellow, I.J.; Pouget-Abadie, J.; Mirza, M.; Xu, B.; Warde-Farley, D.; Ozair, S.; Courville, A.; Bengio, Y. Generative adversarial nets. *Adv. Neural Inf. Process. Syst.* **2014**, *27*, 2672–2680.
48. Douzas, G.; Bação, F. Effective data generation for imbalanced learning using conditional generative adversarial networks. *Expert Syst. Appl.* **2018**, *91*, 464–471. [[CrossRef](#)]
49. Mirza, M.; Osindero, S. Conditional Generative Adversarial Nets. *arXiv* **2014**, arXiv:1411.1784. [[CrossRef](#)]
50. Arjovsky, M.; Chintala, S.; Bottou, L. Wasserstein GAN. *arXiv* **2017**, arXiv:1701.07875. [[PubMed](#)]
51. Arjovsky, M.; Bottou, L. Towards Principled Methods for Training Generative Adversarial Networks. *arXiv* **2017**, arXiv:1701.04862. [[CrossRef](#)]
52. Gulrajani, I.; Ahmed, F.; Arjovsky, M.; Dumoulin, V.; Courville, A.C. Improved training of wasserstein gans. *Adv. Neural Inf. Process. Syst.* **2017**, *30*, 5767–5777.
53. Li, Z.; Zheng, T.; Wang, Y.; Cao, Z.; Guo, Z.; Fu, H. A novel method for imbalanced fault diagnosis of rotating machinery based on generative adversarial networks. *IEEE Trans. Instrum. Meas.* **2020**, *70*, 3500417. [[CrossRef](#)]
54. Wang, C.; Li, F.; Liu, Q.; Wang, H.; Benmoussa, P.; Jeschke, S.; Oeser, M. Establishment and extension of digital aggregate database using auxiliary classifier Wasserstein GAN with gradient penalty. *Constr. Build. Mater.* **2021**, *300*, 124217. [[CrossRef](#)]
55. Luo, J.; Huang, J.; Li, H. A case study of conditional deep convolutional generative adversarial networks in machine fault diagnosis. *J. Intell. Manuf.* **2021**, *32*, 407–425. [[CrossRef](#)]

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.